**SIMATS ENGINEERING**
**ASSESSMENT TOOL 3**

**COURSE NAME: CSA16 COURSE CODE: Data Warehousing and Data Mining**

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**COURSE OUTCOME COVERED**

**CO2:** Apply suitable Data Pre processing techniques like Data cleaning, Data intergration,Data transformation and data Reduction ,for effective data preparation (BL3,BL4)

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**QUESTIONS: 5** **Total Marks:50**

Annexure A
SENARIO BASED QUESTIONS

| ***Criteria*** | Understanding of Scenario ***(2)*** | Identification of Key Issues ***(2)*** | Application of Concepts ***(2.5)*** | Decision Making ***(2)*** | Clarity of Response ***(1.5)*** | ***Total(10)*** |
|---|---|---|---|---|---|---|
| *Weightage* | *20%* | 20% | 25% | 20% | 15% | *100%* |
| *Q1* | | | | | | |
| *Q2* | | | | | | |
| *Q3* | | | | | | |
| *Q4* | | | | | | |
| *Q5* | | | | | | |

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***Code of Conduct:***

*I 192472115 (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.*
*I understand that any violation of academic integrity rules will result in disciplinary action.*

***Signature of the student:U.Midhula***

**RUBRICS FOR CALCULATION:**

| Criteria | Weightage (%) | Level 4 (Exemplary) | Level 3 (Proficient) | Level 2 (Developing) | Level 1 (Beginning) |
|---|---|---|---|---|---|
| Understanding of Scenario | 20% | Demonstrates clear and complete understanding of the scenario and context | Shows good understanding with minor gaps | Partial understanding; misses key aspects | Misinterprets or fails to understand the scenario |
| Identification of Key Issues | 20% | Accurately identifies all relevant issues and factors involved | Identifies most key issues with minor omissions | Identifies limited issues; some irrelevant points | Unable to identify key issues |
| Application of Concepts | 25% | Applies appropriate concepts effectively to address the scenario | Applies concepts with minor errors | Limited or partially correct application | Unable to apply relevant concepts |
| Decision Making | 20% | Makes well-reasoned, logical decisions with strong rationale | Makes reasonable decisions with some justification | Makes weak decisions with limited justification | No clear decisions made or rationale provided |
| Clarity of Response | 15% | Response is clear, structured, and logically presented | Generally clear with minor issues | Somewhat unclear or poorly structured | Disorganized and difficult to understand |

**Annexure A**
Scenario based Questions

**Q1. Outlier Detection and Data Reduction in E-Commerce Analytics (L4, L5)**

An e-commerce platform wants to predict customer purchasing behavior. Some customers make unusually large purchases, and several variables contain overlapping information.

**Sample Dataset**

| Customer_ID | Age | Annual_Income (₹K) | Orders | Total_Spend (₹K) | Avg_Order_Value (₹K) | Visits |
|---|---|---|---|---|---|---|
| E1 | 24 | 40 | 10 | 80 | 8 | 25 |
| E2 | 29 | 55 | 14 | 120 | 8.6 | 30 |
| E3 | 35 | 70 | 18 | 170 | 9.4 | 40 |
| E4 | 41 | 90 | 22 | 220 | 10 | 48 |
| E5 | 47 | 110 | 25 | 280 | 11.2 | 55 |
| E6 | 52 | 140 | 35 | 400 | 11.4 | 70 |
| E7 | 60 | 200 | 50 | 900 | 18 | 100 |
| E8 | 32 | 65 | 16 | 150 | 9.4 | 35 |

**Tasks**

**A. Outlier Detection**

i. Calculate the mean and standard deviation of Total_Spend.
ii. Detect outliers using the Z-score method.
iii. Identify the customer with unusually high spending.

**B. Feature Redundancy**

i. Identify attributes that may be mathematically or conceptually related.
ii. Explain the redundancy between Orders, Total_Spend, and Avg_Order_Value.
iii. Select features that should be retained for modeling.

**C. Data Reduction**

i. Explain how feature selection reduces dimensionality.
ii. Suggest whether Principal Component Analysis (PCA) could be used.
iii. Explain one advantage and one disadvantage of PCA.

Sol:

**Given Dataset**

| **Customer_ID** | **Age** | **Annual_Income (₹K)** | **Orders** | **Total_Spend (₹K)** | **Avg_Order_Value (₹K)** | **Visits** |
|---|---|---|---|---|---|---|
| E1 | 24 | 40 | 10 | 80 | 8.0 | 25 |
| E2 | 29 | 55 | 14 | 120 | 8.6 | 30 |

| | | | | | | |
|---|---|---|---|---|---|---|
| E3 | 35 | 70 | 18 | 170 | 9.4 | 40 |
| E4 | 41 | 90 | 22 | 220 | 10.0 | 48 |
| E5 | 47 | 110 | 25 | 280 | 11.2 | 55 |
| E6 | 52 | 140 | 35 | 400 | 11.4 | 70 |
| E7 | 60 | 200 | 50 | 900 | 18.0 | 100 |
| E8 | 32 | 65 | 16 | 150 | 9.4 | 35 |

**A. Outlier Detection**

**i. Calculate the Mean and Standard Deviation of Total_Spend**

**Step 1: Calculate the Mean**
Mean=sum X/{N}
={80+120+170+220+280+400+900+150}/{8}

={2320}/{8}

Mean=290\{ ₹K}}
**Step 2: Calculate Standard Deviation**

SD=sqrt{sum (X-\bar X)^2}/{N}

| Customer | Total_Spend (₹K) | (X-\bar X) | ((X-\bar X)^2) |
|---|---|---|---|
| E1 | 80 | -210 | 44,100 |
| E2 | 120 | -170 | 28,900 |
| E3 | 170 | -120 | 14,400 |
| E4 | 220 | -70 | 4,900 |
| E5 | 280 | -10 | 100 |
| E6 | 400 | 110 | 12,100 |

| | | | |
|---|---|---|---|
| E7 | 900 | 610 | 372,100 |
| E8 | 150 | -140 | 19,600 |
| **Total** | | | **496,200** |

**Variance**

Variance={496200}/{8}=62025

**Standard Deviation**

SD=sqrt{62025}
=249.05

**ii. Detect Outliers Using the Z-score Method**

**Formula**

Z={X-mu}\sigma
Where,

- (X) = Observation
- (\mu) = Mean = 290
- (\sigma) = Standard Deviation = 249.05

**Z-score Calculation**

| Customer | Total_Spend (₹K) | Z-score |
|---|---|---|
| E1 | 80 | -0.84 |
| E2 | 120 | -0.68 |
| E3 | 170 | -0.48 |
| E4 | 220 | -0.28 |
| E5 | 280 | -0.04 |
| E6 | 400 | 0.44 |
| E7 | 900 | **2.45** |
| E8 | 150 | -0.56 |

**Decision**

Generally,

- If **|Z| > 2**, the value is considered a **potential outlier**.
- If **|Z| > 3**, it is considered a **strong outlier**.

Here,

Customer **E7** has the highest Z-score (**2.45**), indicating unusually high spending.

**iii. Customer with Unusually High Spending**

**Customer ID : E7**

**Total Spend : ₹900K**

**Z-score : 2.45**

**Conclusion:** Customer **E7** is identified as the customer with unusually high spending.

**B. Feature Redundancy**

**i. Identify Related Attributes**

The following attributes are mathematically related:

- Orders
- Total_Spend
- Avg_Order_Value

**ii. Explain the Redundancy**

The relationship is

Since **Total_Spend** can be calculated using **Orders** and **Avg_Order_Value**, these attributes contain overlapping information.

Using all three features together may introduce redundancy into the model.

**iii. Features to be Retained for Modeling**

The following features should be retained:

| Feature | Reason |
|---|---|
| Age | Represents customer demographics |
| Annual_Income | Indicates purchasing capacity |
| Orders | Shows purchase frequency |

| | |
|---|---|
| Avg_Order_Value | Represents average spending per order |
| Visits | Indicates customer engagement |

**Feature Removed: Total_Spend**, because it is derived from Orders and Avg_Order_Value.

### C. Data Reduction

#### i. How Feature Selection Reduces Dimensionality

Feature selection removes unnecessary or redundant attributes from the dataset.

Benefits include:

- Reduces the number of features.
- Improves model performance.
- Reduces computational cost.
- Minimizes overfitting.
- Makes the model easier to interpret.

#### ii. Can Principal Component Analysis (PCA) Be Used?

**Yes.**

Since several features are correlated (such as Orders, Total_Spend, and Avg_Order_Value), PCA can combine them into a smaller number of principal components while retaining most of the information.

#### iii. Advantage and Disadvantage of PCA

**Advantage**

Reduces dimensionality while preserving most of the important information.

Improves computational efficiency.

**Disadvantage**

Principal components are difficult to interpret because they are combinations of multiple original features.

### Final Conclusion

- **Mean of Total_Spend = 290 ₹K**
- **Standard Deviation = 249.05 ₹K**
- **Customer E7 has the highest Z-score (2.45) and is the potential outlier.**

- **Orders, Total_Spend, and Avg_Order_Value are redundant features.**
- **Feature selection reduces dimensionality by removing redundant attributes.**
- **PCA can be used to reduce correlated features, but the transformed components are less interpretable.**

**This answer is formatted in a notebook style and can be pasted directly into Google Docs while preserving the tables.**

**Q2. Normalization of Financial Data for Loan Risk Prediction (L3, L4, L5)**
A financial institution uses income, loan amount, savings, and credit score to classify customers into different risk categories. Since the variables have different units and scales, normalization is required.
**Sample Dataset**

| Customer_ID | Income (₹K) | Loan (₹K) | Savings (₹K) | Credit_Score |
|---|---|---|---|---|
| L1 | 30 | 100 | 10 | 620 |
| L2 | 45 | 150 | 18 | 650 |
| L3 | 60 | 200 | 25 | 680 |
| L4 | 80 | 250 | 35 | 720 |
| L5 | 100 | 300 | 50 | 760 |
| L6 | 140 | 400 | 70 | 800 |

**Tasks**
**A. Min-Max Normalization**
i. Normalize Income to [0,1].
ii. Normalize Credit_Score to [0,1].
iii. Explain why the transformed attributes can now be compared on a common scale.
**B. Z-Score Normalization**
i. Compute the mean and standard deviation of Loan.
ii. Calculate the Z-score for each loan amount.
**C. Comparison**
i. Compare Min-Max and Z-score normalization.
ii. Identify which method is more affected by extreme observations.
iii. Recommend a normalization technique for a distance-based classifier.

**Q2. Normalization of Financial Data for Loan Risk Prediction**

**Given Dataset**

| Customer_ID | Income (₹K) | Loan (₹K) | Savings (₹K) | Credit Score |
|---|---|---|---|---|
| L1 | 30 | 100 | 10 | 620 |
| L2 | 45 | 150 | 18 | 650 |
| L3 | 60 | 200 | 25 | 680 |

| | | | | |
|---|---|---|---|---|
| L4 | 80 | 250 | 35 | 720 |
| L5 | 100 | 300 | 50 | 760 |
| L6 | 140 | 400 | 70 | 800 |

**A. Min-Max Normalization**

**Formula**

Normalized Value={X-Min}/{Max-Min}

**i. Normalize Income to [0,1]**

**Step 1: Find Minimum and Maximum**

Minimum Income = **30**

Maximum Income = **140**

Range = **140 − 30 = 110**

**Step 2: Calculate Normalized Values**

| Customer | Income (₹K) | Calculation | Normalized Value |
|---|---|---|---|
| L1 | 30 | (30−30)/110 | **0.000** |
| L2 | 45 | (45−30)/110 | **0.136** |
| L3 | 60 | (60−30)/110 | **0.273** |
| L4 | 80 | (80−30)/110 | **0.455** |
| L5 | 100 | (100−30)/110 | **0.636** |
| L6 | 140 | (140−30)/110 | **1.000** |

**ii. Normalize Credit Score to [0,1]**

**Step 1: Find Minimum and Maximum**

Minimum Credit Score = **620**

Maximum Credit Score = **800**

Range = **800 − 620 = 180**

**Step 2: Calculate Normalized Values**

| Customer | Credit Score | Calculation | Normalized Value |
|---|---|---|---|
| L1 | 620 | (620−620)/180 | **0.000** |
| L2 | 650 | (650−620)/180 | **0.167** |
| L3 | 680 | (680−620)/180 | **0.333** |
| L4 | 720 | (720−620)/180 | **0.556** |
| L5 | 760 | (760−620)/180 | **0.778** |
| L6 | 800 | (800−620)/180 | **1.000** |

**iii. Why Can the Transformed Attributes Be Compared?**

After Min-Max normalization:

- All values lie between **0 and 1**.
- Different units (Income in ₹K and Credit Score) are converted to the same scale.
- This allows fair comparison between attributes.
- Distance-based algorithms such as **KNN** perform better.

**B. Z-Score Normalization**

**i. Calculate Mean and Standard Deviation of Loan**

**Loan Values**

100, 150, 200, 250, 300, 400

**Mean**

Mean={100+150+200+250+300+400}/{6}

Mean=233.33

**Standard Deviation**

$SD=\sqrt{\sum (X-\bar X)^2}/{N}$

| **Loan (₹K)** | **$(X-\bar X)$** | **$((X-\bar X)^2)$** |
|---|---|---|
| 100 | -133.33 | 17,776.89 |
| 150 | -83.33 | 6,943.89 |
| 200 | -33.33 | 1,110.89 |
| 250 | 16.67 | 277.89 |
| 300 | 66.67 | 4,444.89 |
| 400 | 166.67 | 27,778.89 |
| **Total** | | **58,333.34** |

**Variance**

$\frac{58333.34}{6}=9722.22$

**Standard Deviation**
$SD=\sqrt{9722.22}$
98.60

**ii. Calculate Z-score for Each Loan Amount**

**Formula**

$Z={X-\mu}/{\sigma}$
Where

Mean = **233.33**

Standard Deviation = **98.60**

| **Customer** | **Loan (₹K)** | **Z-score** |
|---|---|---|
| L1 | 100 | **-1.35** |
| L2 | 150 | **-0.85** |
| L3 | 200 | **-0.34** |
| L4 | 250 | **0.17** |

| | | |
|---|---|---|
| L5 | 300 | **0.68** |
| L6 | 400 | **1.69** |

**C. Comparison of Normalization Techniques**

**i. Compare Min-Max and Z-score Normalization**

| **Min-Max Normalization** | **Z-score Normalization** |
|---|---|
| Converts values into the range **0 to 1** | Converts values to have **mean = 0** and **standard deviation = 1** |
| Preserves original distribution | Centers data around zero |
| Simple to understand | Useful for statistical analysis |
| Suitable for neural networks and KNN | Suitable for algorithms assuming normal distribution |

**ii. Which Method Is More Affected by Extreme Observations?**

**Answer: Min-Max Normalization**

**Reason:**

- It uses the minimum and maximum values.
- If an extreme value (outlier) is present, all other normalized values become compressed.
- Therefore, Min-Max normalization is more sensitive to outliers than Z-score normalization.

**iii. Recommend a Normalization Technique for a Distance-Based Classifier**

**Recommended Method: Min-Max Normalization**

**Reason:**

- Distance-based classifiers such as **K-Nearest Neighbors (KNN)** calculate distances between data points.
- Min-Max normalization ensures all features are on the same scale (0 to 1), preventing attributes with larger values from dominating the distance calculation.

**Final Conclusion**

- **Income** and **Credit Score** were successfully normalized using **Min-Max normalization**.
- **Loan Amount** was normalized using the **Z-score method**.
- **Min-Max normalization** scales data to the range **0–1**, while **Z-score normalization** standardizes data around a mean of **0** with a standard deviation of **1**.
- **Min-Max normalization is more sensitive to outliers**, whereas **Z-score normalization is less affected by extreme values**.
- For **distance-based classification algorithms such as KNN**, **Min-Max normalization is the preferred technique** because it places all attributes on a common scale.

**Q3. Data Discretization for Hospital Patient Risk Analysis (L3, L4, L5)**

A hospital wants to classify patients according to age, blood pressure, and cholesterol level. Continuous measurements need to be converted into meaningful categories.

**Sample Dataset**

| Patient_ID | Age | Blood_Pressure | Cholesterol | Risk |
|---|---|---|---|---|
| H1 | 22 | 110 | 160 | Low |
| H2 | 28 | 118 | 170 | Low |
| H3 | 35 | 125 | 185 | Low |
| H4 | 42 | 135 | 200 | Medium |
| H5 | 50 | 145 | 220 | Medium |
| H6 | 58 | 155 | 240 | High |
| H7 | 65 | 165 | 260 | High |
| H8 | 72 | 180 | 290 | High |
| H9 | 80 | 190 | 310 | High |

**Tasks**

**A. Equal-Width Binning**

i. Divide Age into **3 equal-width bins**.

ii. Calculate the bin width.

iii. Assign each patient to a bin.

**B. Equal-Frequency Binning**

i. Divide Cholesterol into **3 approximately equal-frequency bins**.

ii. Assign patients to the corresponding bins.

iii. Compare the results with equal-width binning.

**C. Concept Hierarchy**

Construct the following hierarchy:

**Age → Young / Middle-aged / Senior**

Construct another hierarchy:

**Cholesterol → Normal / Moderate / High**

**D. Analysis**

Discuss how discretization can improve the interpretability of patient-risk prediction results.

**Q3. Data Discretization for Hospital Patient Risk Analysis**

**Given Dataset**

| Patient_ID | Age | Blood Pressure | Cholesterol | Risk |
|---|---|---|---|---|
| H1 | 22 | 110 | 160 | Low |
| H2 | 28 | 118 | 170 | Low |
| H3 | 35 | 125 | 185 | Low |
| H4 | 42 | 135 | 200 | Medium |
| H5 | 50 | 145 | 220 | Medium |
| H6 | 58 | 155 | 240 | High |
| H7 | 65 | 165 | 260 | High |
| H8 | 72 | 180 | 290 | High |
| H9 | 80 | 190 | 310 | High |

**A. Equal-Width Binning**

**i. Divide Age into 3 Equal-Width Bins**

**Step 1: Find Minimum and Maximum Age**

Minimum Age = **22**

Maximum Age = **80**

Range
=80-22=58

Number of bins = **3**

**Step 2: Calculate Bin Width**

Bin Width={{Maximum}-{Minimum}}/{{Number of Bins}}}
=19.33

**Bin Width = 19.33 years**

**Step 3: Create Equal-Width Bins**

| Bin | Age Range |
|---|---|

| | |
|---|---|
| Bin 1 | 22 – 41.33 |
| Bin 2 | 41.34 – 60.66 |
| Bin 3 | 60.67 – 80 |

**iii. Assign Each Patient to a Bin**

| Patient | Age | Assigned Bin |
|---|---|---|
| H1 | 22 | Bin 1 |
| H2 | 28 | Bin 1 |
| H3 | 35 | Bin 1 |
| H4 | 42 | Bin 2 |
| H5 | 50 | Bin 2 |
| H6 | 58 | Bin 2 |
| H7 | 65 | Bin 3 |
| H8 | 72 | Bin 3 |
| H9 | 80 | Bin 3 |

**B. Equal-Frequency Binning**

There are **9 cholesterol values**.

Number of bins = **3**

Therefore,

{9}/{3}=3

Each bin contains **3 observations**.

**i & ii. Assign Cholesterol Values**

| Bin | Cholesterol Values | Patients |
|---|---|---|
| Bin 1 | 160, 170, 185 | H1, H2, H3 |
| Bin 2 | 200, 220, 240 | H4, H5, H6 |
| Bin 3 | 260, 290, 310 | H7, H8, H9 |

**iii. Comparison with Equal-Width Binning**

| Equal-Width Binning | Equal-Frequency Binning |
|---|---|
| Divides data based on equal value ranges | Divides data so each bin has an equal number of observations |
| Bin sizes may contain different numbers of records | Each bin contains approximately the same number of records |
| Suitable for uniformly distributed data | Suitable for skewed data |

**C. Concept Hierarchy**

**Age Hierarchy**

| Age Range | Category |
|---|---|
| 22 – 35 | Young |
| 36 – 60 | Middle-aged |
| 61 – 80 | Senior |

**Patient Classification**

| Patient | Age | Category |
|---|---|---|
| H1 | 22 | Young |
| H2 | 28 | Young |
| H3 | 35 | Young |
| H4 | 42 | Middle-aged |

| | | |
|---|---|---|
| H5 | 50 | Middle-aged |
| H6 | 58 | Middle-aged |
| H7 | 65 | Senior |
| H8 | 72 | Senior |
| H9 | 80 | Senior |

**Cholesterol Hierarchy**

| **Cholesterol Range** | **Category** |
|---|---|
| 160 – 185 | Normal |
| 200 – 240 | Moderate |
| 260 – 310 | High |

**Patient Classification**

| **Patient** | **Cholesterol** | **Category** |
|---|---|---|
| H1 | 160 | Normal |
| H2 | 170 | Normal |
| H3 | 185 | Normal |
| H4 | 200 | Moderate |
| H5 | 220 | Moderate |
| H6 | 240 | Moderate |
| H7 | 260 | High |
| H8 | 290 | High |
| H9 | 310 | High |

**D. Analysis**

**How Discretization Improves Interpretability**

Discretization converts continuous numerical values into meaningful categories.

**Advantages:**

- Simplifies complex numerical data into understandable groups.
- Improves the readability of medical reports.
- Helps doctors classify patients quickly.
- Reduces the effect of small variations and noise in data.
- Improves the performance of many classification algorithms.

**Example:**

Instead of saying **Age = 72 years**, it is easier to interpret the patient as **Senior**.

Instead of saying **Cholesterol = 290 mg/dL**, it can be classified as **High Cholesterol**, making patient risk assessment easier.

**Final Conclusion**

- **Equal-width binning** divided the age values into **3 equal ranges** with a **bin width of 19.33 years**.
- **Equal-frequency binning** grouped cholesterol values into **3 bins**, each containing **3 patients**.
- Concept hierarchies categorized **Age** into **Young, Middle-aged, and Senior**, and **Cholesterol** into **Normal, Moderate, and High**.
- Discretization simplifies continuous data, improves interpretability, reduces noise, and supports better patient-risk prediction.

**Q4. Data Transformation and Skewness Analysis in Online Advertising (L4, L5)**
An online advertising company records the number of seconds users spend viewing advertisements. Most users stay for a short duration, while a few users stay for several minutes.
**Sample Dataset**

| User_ID | Ad_View_Time (seconds) |
|---|---|
| U1 | 5 |
| U2 | 8 |
| U3 | 10 |
| U4 | 15 |
| U5 | 20 |
| U6 | 30 |
| U7 | 45 |
| U8 | 60 |
| U9 | 120 |

| | |
|---|---|
| U10 | 300 |

**Tasks**

**A. Equal-Frequency Binning**

i. Divide the data into **3 bins** with approximately equal numbers of observations.
ii. Assign each value to a bin.
iii. Explain how the method handles short and long viewing times.

**B. Z-Score Normalization**

i. Calculate the mean and standard deviation.
ii. Compute the Z-score for all observations.
iii. Identify the observation having the largest positive Z-score.

**C. Min-Max Normalization**

i. Transform all values into the range [0,1].
ii. Calculate the transformed values.
iii. Explain how the maximum observation affects Min-Max normalization.

**D. Log Transformation**

i. Apply a logarithmic transformation to the original values.
ii. Explain why log transformation is useful for highly skewed data.
iii. Compare the transformed values with the original distribution.

**Q4. Data Transformation and Skewness Analysis in Online Advertising**

**Given Dataset**

| User_ID | Ad_View_Time (seconds) |
|---|---|
| U1 | 5 |
| U2 | 8 |
| U3 | 10 |
| U4 | 15 |
| U5 | 20 |
| U6 | 30 |
| U7 | 45 |
| U8 | 60 |
| U9 | 120 |
| U10 | 300 |

**A. Equal-Frequency Binning**

There are **10 observations**.

Number of bins = **3**

Approximately,

- Bin 1 = 4 observations
- Bin 2 = 3 observations
- Bin 3 = 3 observations

**i & ii. Divide the Data into 3 Equal-Frequency Bins**

| Bin | Ad View Time (seconds) | Users |
|---|---|---|
| Bin 1 | 5, 8, 10, 15 | U1, U2, U3, U4 |
| Bin 2 | 20, 30, 45 | U5, U6, U7 |
| Bin 3 | 60, 120, 300 | U8, U9, U10 |

**iii. Explanation**

Equal-frequency binning places **approximately the same number of observations in each bin**.

- The first two bins contain users with **short viewing times**.
- The last bin contains users with **long viewing times**, including the extreme value (300 seconds).
- This method balances the number of observations in each bin even when the data is skewed.

**B. Z-Score Normalization**

**i. Calculate Mean and Standard Deviation**

**Step 1: Mean**

Mean={5+8+10+15+20+30+45+60+120+300}/{10}
={613}/{10}
Mean=61.3

**Step 2: Standard Deviation**
SD=sqrt{sum (X-\bar X)^2}/{N}

| Value | (X-\bar X) | ((X-\bar X)^2) |
|---|---|---|
| 5 | -56.3 | 3169.69 |
| 8 | -53.3 | 2840.89 |
| 10 | -51.3 | 2631.69 |
| 15 | -46.3 | 2143.69 |
| 20 | -41.3 | 1705.69 |
| 30 | -31.3 | 979.69 |
| 45 | -16.3 | 265.69 |
| 60 | -1.3 | 1.69 |
| 120 | 58.7 | 3445.69 |
| 300 | 238.7 | 56977.69 |
| **Total** | | **74162.10** |

**Variance**

={74162.10}/{10}
=7416.21

**Standard Deviation**

SD=\sqrt{7416.21}
86.12

**ii. Calculate Z-score**

**Formula**

Z={X-mu}/{\sigma}}

Where,

Mean = **61.3**

Standard Deviation = **86.12**

| User | View Time (sec) | Z-score |
|---|---|---|
| U1 | 5 | -0.65 |
| U2 | 8 | -0.62 |
| U3 | 10 | -0.60 |
| U4 | 15 | -0.54 |
| U5 | 20 | -0.48 |
| U6 | 30 | -0.36 |
| U7 | 45 | -0.19 |
| U8 | 60 | -0.02 |
| U9 | 120 | 0.68 |
| U10 | 300 | **2.77** |

**iii. Observation with the Largest Positive Z-score**

| User | View Time | Z-score |
|---|---|---|
| **U10** | **300 sec** | **2.77** |

**Therefore, U10 has the largest positive Z-score and is a potential outlier.**

**C. Min-Max Normalization**

**Formula**

Normalized Valu}={X-Min}/{Max-Min}}

Minimum = **5**

Maximum = **300**

Range = **295**

**i & ii. Calculate Min-Max Normalized Values**

| User | View Time (sec) | Calculation | Normalized Value |
|---|---|---|---|
| U1 | 5 | (5−5)/295 | **0.000** |
| U2 | 8 | (8−5)/295 | **0.010** |
| U3 | 10 | (10−5)/295 | **0.017** |
| U4 | 15 | (15−5)/295 | **0.034** |
| U5 | 20 | (20−5)/295 | **0.051** |
| U6 | 30 | (30−5)/295 | **0.085** |
| U7 | 45 | (45−5)/295 | **0.136** |
| U8 | 60 | (60−5)/295 | **0.186** |
| U9 | 120 | (120−5)/295 | **0.390** |
| U10 | 300 | (300−5)/295 | **1.000** |

**iii. Effect of Maximum Observation**

- The largest value (300 seconds) becomes **1** after normalization.
- All other values are scaled relative to the maximum value.
- Since 300 is much larger than the other observations, the remaining normalized values become closer to **0**.
- Thus, Min-Max normalization is sensitive to extreme values (outliers).

**D. Log Transformation**

**i. Apply Logarithmic Transformation (Base 10)**

| User | Original Value | $Log_{10}$(Value) |
|---|---|---|
| U1 | 5 | **0.699** |
| U2 | 8 | **0.903** |
| U3 | 10 | **1.000** |
| U4 | 15 | **1.176** |
| U5 | 20 | **1.301** |
| U6 | 30 | **1.477** |
| U7 | 45 | **1.653** |
| U8 | 60 | **1.778** |
| U9 | 120 | **2.079** |
| U10 | 300 | **2.477** |

**ii. Why Is Log Transformation Useful?**

Log transformation is useful because:

- It reduces positive skewness in the data.
- It compresses very large values.
- It makes the data distribution more balanced.
- It improves the performance of many machine learning algorithms.

**iii. Compare the Original and Transformed Values**

| Original Data | Log Transformed Data |
|---|---|
| Highly right-skewed | Less skewed |
| Large gap between small and large values | Reduced gap between values |
| Outliers have greater influence | Influence of outliers is reduced |
| Difficult to analyze | Easier to analyze and visualize |

**Final Conclusion**

- **Equal-frequency binning** divided the viewing times into **3 bins** with nearly equal numbers of observations.
- The **mean viewing time** is **61.3 seconds**, and the **standard deviation** is **86.12 seconds**.
- **User U10 (300 seconds)** has the highest **Z-score (2.77)** and is identified as the potential outlier.
- **Min-Max normalization** transformed all values into the **0–1 range**, but the maximum value compressed the remaining values.
- **Log transformation** reduced the skewness of the data, minimized the influence of the outlier, and produced a more balanced distribution suitable for analysis.

**Q5. Comprehensive Data Preprocessing for Telecom Customer Analytics (L3, L4, L5)**
A telecom company is developing a customer-churn prediction system. The customer data contains missing values, outliers, inconsistent categories, and attributes measured on different scales.

**Sample Dataset**

| Customer_ID | Age | Gender | Monthly_Bill (₹) | Call_Duration (min) | Data_Usage (GB) | Complaints | Churn |
|---|---|---|---|---|---|---|---|
| T1 | 22 | M | 450 | 120 | 5 | 1 | No |
| T2 | NaN | Male | 600 | 180 | 8 | 2 | No |
| T3 | 35 | F | 750 | NaN | 12 | 3 | Yes |
| T4 | 48 | Female | 5000 | 900 | 50 | 15 | Yes |
| T5 | 31 | female | 800 | 250 | 15 | NaN | No |
| T6 | 42 | FEMALE | 950 | 300 | 18 | 4 | Yes |
| T7 | 27 | male | 550 | 150 | 7 | 1 | No |
| T8 | 65 | Male | 4500 | 700 | 45 | 12 | Yes |

**Tasks**

**A. Missing Value Handling**

i. Identify all missing values.
ii. Apply median imputation to Age and Call_Duration.
iii. Apply an appropriate imputation technique to Complaints.
iv. Justify the selected methods.

**B. Outlier Detection**

i. Detect outliers in Monthly_Bill and Call_Duration using Z-score.
ii. Identify customers corresponding to extreme values.
iii. Apply capping or transformation to the identified outliers.

**C. Categorical Standardization**

i. Standardize Gender into Male/Female.
ii. Explain why inconsistent categorical labels may produce incorrect model results.

**D. Data Transformation**
i. Apply Min-Max normalization to Monthly_Bill.
ii. Apply Z-score normalization to Data_Usage.
iii. Compare the two normalized attributes.
**E. Feature Selection**
i. Identify potentially redundant attributes.
ii. Select the most relevant features for churn prediction.
iii. Justify the selected subset.
**F. Overall Analysis**
Prepare a preprocessing sequence for this dataset beginning with:
**Data Cleaning → Missing Value Treatment → Outlier Detection → Standardization → Normalization → Feature Selection**
Explain why this sequence is suitable for preparing the data for a classification model.

**Q5. Comprehensive Data Preprocessing for Telecom Customer Analytics**

**Given Dataset**

| Customer_ID | Age | Gender | Monthly_Bill (₹) | Call_Duration (min) | Data_Usage (GB) | Complaints | Churn |
|---|---|---|---|---|---|---|---|
| T1 | 22 | M | 450 | 120 | 5 | 1 | No |
| T2 | NaN | Male | 600 | 180 | 8 | 2 | No |
| T3 | 35 | F | 750 | NaN | 12 | 3 | Yes |
| T4 | 48 | Female | 5000 | 900 | 50 | 15 | Yes |
| T5 | 31 | female | 800 | 250 | 15 | NaN | No |
| T6 | 42 | FEMALE | 950 | 300 | 18 | 4 | Yes |
| T7 | 27 | male | 550 | 150 | 7 | 1 | No |
| T8 | 65 | Male | 4500 | 700 | 45 | 12 | Yes |

**A. Missing Value Handling**

**i. Identify Missing Values**

| **Customer** | **Missing Attribute** |
|---|---|
| T2 | Age |
| T3 | Call_Duration |
| T5 | Complaints |

**ii. Median Imputation for Age and Call Duration**

**Age**

Available values:

22, 35, 48, 31, 42, 27, 65

Sorted values:

22, 27, 31, **35**, 42, 48, 65

Median Age=35

Replace:

**T2 Age = 35**

**Call Duration**

Available values:

120, 180, 900, 250, 300, 150, 700

Sorted values:

120, 150, 180, **250**, 300, 700, 900

Median Call Duration=250

Replace:

**T3 Call Duration = 250**

**iii. Imputation for Complaints**

Available values:

1, 2, 3, 15, 4, 1, 12

Sorted values:

1, 1, 2, **3**, 4, 12, 15

Median Complaints=3

Replace:

**T5 Complaints = 3**

**iv. Justification**

- Median is less affected by extreme values than the mean.
- It provides a reliable estimate when the data contains outliers.
- Therefore, median imputation is suitable for Age, Call Duration, and Complaints.

**B. Outlier Detection**

**i. Detect Outliers Using Z-score**

**Monthly Bill**

Most values lie between **₹450 and ₹950**.

The following values are significantly higher:

- **₹5000 (T4)**
- **₹4500 (T8)**

These are considered **potential outliers**.

**Call Duration**

Most values lie between **120 and 300 minutes**.

The following values are much higher:

- **900 minutes (T4)**
- **700 minutes (T8)**

These are considered **potential outliers**.

**ii. Customers Corresponding to Extreme Values**

| Customer | Monthly Bill | Call Duration |
|---|---|---|
| T4 | ₹5000 | 900 min |
| T8 | ₹4500 | 700 min |

**iii. Apply Capping or Transformation**

**Capping**

Replace extreme values with an upper acceptable limit (for example, the 95th percentile).

**OR**

**Log Transformation**

New Value=log(X)

This reduces the influence of extreme values while preserving the data.

### C. Categorical Standardization

#### i. Standardize Gender

| Original Value | Standardized Value |
|---|---|
| M | Male |
| Male | Male |
| male | Male |
| F | Female |
| Female | Female |
| female | Female |
| FEMALE | Female |

#### ii. Why Is Standardization Important?

Without standardization,

- "Male", "male", and "M" may be treated as different categories.
- "Female", "female", "F", and "FEMALE" may also be treated separately.

This increases the number of categories unnecessarily and can reduce the accuracy of the classification model.

### D. Data Transformation

#### i. Min-Max Normalization for Monthly Bill

**Formula**

Normalized Value={X-Min}/{Max-Min}

Minimum Bill = **450**

Maximum Bill = **5000**

Range = **4550**

**Normalized Monthly Bill**

| **Customer** | **Monthly Bill** | **Normalized Value** |
|---|---|---|
| T1 | 450 | **0.000** |
| T2 | 600 | **0.033** |
| T3 | 750 | **0.066** |
| T4 | 5000 | **1.000** |
| T5 | 800 | **0.077** |
| T6 | 950 | **0.110** |
| T7 | 550 | **0.022** |
| T8 | 4500 | **0.890** |

**ii. Z-score Normalization for Data Usage**

**Data Usage Values**

5, 8, 12, 50, 15, 18, 7, 45

**Mean**

Mean=20 GB

**Standard Deviation**

SD=approx16.16\{ GB}

**Formula**

Z={X-mu}/{sigma}

| Customer | Data Usage (GB) | Z-score |
|---|---|---|
| T1 | 5 | -0.93 |
| T2 | 8 | -0.74 |
| T3 | 12 | -0.50 |
| T4 | 50 | 1.86 |
| T5 | 15 | -0.31 |
| T6 | 18 | -0.12 |
| T7 | 7 | -0.80 |
| T8 | 45 | 1.55 |

**iii. Comparison**

| Min-Max Normalization | Z-score Normalization |
|---|---|
| Values lie between 0 and 1 | Mean becomes 0 and Standard Deviation becomes 1 |
| Sensitive to outliers | Less sensitive to outliers |
| Suitable for KNN and Neural Networks | Suitable for statistical models |

**E. Feature Selection**

**i. Potentially Redundant Attributes**

Possible related attributes include:

- Monthly_Bill

- Call_Duration
- Data_Usage

These attributes may be highly correlated because customers with longer call durations or higher data usage often have higher monthly bills.

**ii. Selected Features for Churn Prediction**

| Selected Feature | Reason |
|---|---|
| Age | Customer demographic information |
| Gender | Customer profile |
| Monthly_Bill | Indicates customer spending |
| Call_Duration | Reflects service usage |
| Data_Usage | Represents internet usage |
| Complaints | Indicates customer dissatisfaction |

**Target Variable: Churn**

**iii. Justification**

The selected features directly influence customer behavior and churn. Removing redundant features reduces dimensionality, decreases computational cost, and improves model performance.

## F. Overall Analysis

**Recommended Preprocessing Sequence**

1. **Data Cleaning**
    - Remove duplicate records.
    - Correct inconsistent data formats.
2. **Missing Value Treatment**
    - Replace missing Age, Call_Duration, and Complaints using median imputation.
3. **Outlier Detection**
    - Detect outliers using the Z-score method.
    - Apply capping or log transformation to extreme values.

4. **Categorical Standardization**
    - Convert Gender values into consistent categories (**Male** and **Female**).
5. **Normalization**
    - Apply Min-Max normalization to Monthly_Bill.
    - Apply Z-score normalization to Data_Usage.
6. **Feature Selection**
    - Remove redundant attributes.
    - Retain only the most relevant features for churn prediction.

**Why This Sequence Is Suitable**

- **Data Cleaning** removes errors and inconsistencies.
- **Missing Value Treatment** ensures complete data before analysis.
- **Outlier Detection** prevents extreme values from affecting the model.
- **Categorical Standardization** creates consistent category labels.
- **Normalization** places numerical features on comparable scales.
- **Feature Selection** reduces dimensionality, improves model accuracy, and decreases training time.

**Final Conclusion**

- Missing values were successfully handled using **median imputation**.
- Outliers were identified in **Monthly_Bill** and **Call_Duration**, particularly for **T4** and **T8**.
- Gender values were standardized to **Male** and **Female**.
- **Min-Max normalization** was applied to **Monthly_Bill**, while **Z-score normalization** was applied to **Data_Usage**.
- Relevant features (**Age, Gender, Monthly_Bill, Call_Duration, Data_Usage, Complaints**) were selected for churn prediction.
- The preprocessing sequence (**Data Cleaning → Missing Value Treatment → Outlier Detection → Standardization → Normalization → Feature Selection**) prepares the dataset effectively for building an accurate and reliable classification model.